GENE

Gene

↳ William Johnson - 20k - 25k → Genes in normal humanbeing

↳ Study of gene → Genetics → father → Sir Gregan Mendal.

It determines:

- Skin colour
- diseases
- eye colour

Mitochondria

↳ Powerhouse of the cell
↳ Semi - Autonomous body
↳ cell with in cell
~~↳ कौलिकर insect~~
↳ कौलिकर describe about the mitochondria in insect

↳ ALTMAN & FLEMMING → 1890-91

ALTMAN named → Bioplast

FLEMMING named ↳ Filla

Carl Benda named it as mitochondria.

Work of mitochondria

↳ cellular respiration

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$
↓
+ ATP
↓
अणु
↓
38 ATP
↓
2 ATP - O_2 ✗

ATP → Adinosine Tri Phosphate

ADP → Adinosine Di Phosphate

Lysosomes :-

- लाइसोसोम डी डूवे discovered (1954 - 1955)
- Suicidal Bag
- Atom Bomb
- Disposal Bag
- ~~Suppy कर्मचारी~~

Because

Hydrolytic Enzyme is present.

(4.5 - 5.5) (Acidic)

- खुद की ही कोशिका को खा जाता है
- It fights with germs and bacteria

Unwanted substance

↳ Bacteria

↳ micro - organism $\boxed{50 - 1000}$ Lisosome present in one cell

Lisosome present in ↓

1. WBC
- Sliva
- Kupffer - cell
- Pancreas
- kidney
- Spleen
- osteoclast

Ribosomes

⇒ protein factory

⇒ Ribo - Some > r-RNA + protein

⇒ Size = 150 - 230 Å

Mammals ⇒ 5-10 million Ribo

* Ribosome was first discovered by Robinson & Brown in (Plant cell.)

• ~~Then~~ early Pallad ~~discovered~~ named it as Pallad Praticle.

⇒ Smallest cell organelles

⇒ No membrane

⇒ Smellest कोशिकांग

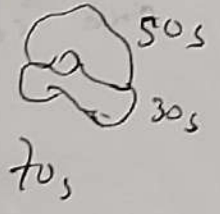
⇒ Helps in protien making

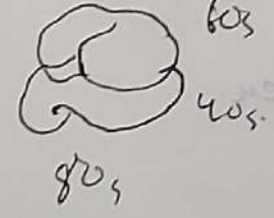
⇒ universal component of cell

RBC → Nucleus, Ribosomes, Lysosomes, Golgi-apparatus
↳ mammals.

Type

70s	80s	Sz sweet meg unit
Prokaryotic	Eukaryotic	
Becteria	↙ ↘	
Chloroplast, mitochondria.	Plants - Animals	
(50s + 30s)	(60s + 40s)	





Vocoules

- Blank space in cell is known as vacules
- discovered and named by ल्यूवेनहॉक
- Pocket of cell
- It absorb extra carbon + water + Glucose

✱ अमीबा अपना भोजन Vocoules के Through पचाता है

Golgi apparatus

↳ 1897 - 1898
↳ It was found in owl cell
↳ कैमिलो गॉल्जी The man who discovered
↳ In Plant cell it is known as Dictiosomes

Its working

- It collect the protien in cell
- also known as trafic police of the cell

⇒ When golgi breaks it makes Lysosomes.

⇒ Sperm cell Tip Acrosomes → Golgi Apparatus.

E.R

Endoplasmic Recticulum (ER)

Discoverer :-

↳ G.H.A. Carlier
↳ Porter → E.R. नामकरण

Endoplasmic Reticulum

SER (Smooth Endoplasmic Reticulum)
↓
Lipids (made with)

RER (Rough Endoplasmic Reticulum)
↓
(made with) Ribosomes

Plastids

↳ Pant cells
↳ discoverer → हेकेले
↳ Named by → A.F.W schimper.

- Size (1-10 micro)
- सबसे dsT कोशिकांग
- कोशिका का रसोईघर

Plant cell (Euglina)

- Chloroplast
  - प्रकाश संश्लेषण
  - Green colour
  - Chlorophyll mg
- Chromoplast
  - Red, Blue
  - Yellow
- Leucoplast
  - White, Brown in the roots

- Green mango → पिले रंग का आम
Chloroplast होता है जब पक जाता है Chromoplast होता है,

Leucoplast :-

- ⇒ Amyloplast - Starch (आलू)
- ⇒ Elioplast - मूंगफली (सरसो)
- ↳ Protienoplast / Aleuroplast ⇒ Protien

↳ This stores carbohydrate
↳ " " Fat
↳ " " Protien

Cell organelles which have membrane

Cell organelles

Single membrane	double membrane	No membrane
Golgibody	Mitochondria	Ribosome
E.R.	chloroplast	Centrosome
Lysosome	Nucleus	microtubules
vacules		

Key points

- RBC dont have Nucleus, Ribosomes Lysosomes, Golgi
- World most smallest cell - mycoplasma plant
  It is also known as Joker of Animal kingdom

other name.

PPLO → poly Pleuro phneumonia Like organism

MLO → molecule Like organism

- Largest cell ⇒ Egg of ostrich
- मनुष्य के शरीर की सबसे लम्बी कोशिका - ~~न्यूरान~~ Neurone
- मनुष्य के शरीर की सबसे बड़ी जनन कोशिका - Ova
  (Gonad cell)
- " " " " छोटी " " - Sperm
  smallest Gonad cell

Disease

Eupenea - Normal

Hyponea → 10↓ {10↓ से कम oxygen lete है तो}

Hyperpnea → 15↑ {15↓ " " " }

Hypoxia → O_2↓ {बहुत कम Oxygen " "}

Anoxia → O_2 (x)

Emphysema → Smoke → Alveolies → Pores (फेफड़े मे धुआ जम जाता है)

Pneumonia → O_2↓ {Bacteria से होता है}

Asthuma → Fungi → R.S.

Nasal cavity

↳ Pollen grains

↳ जुकाम, बुखार

T. B → Tuberculosis

↳ Bacteria (MTB)

BCG → Vaccine {used for Tubercolosis}

White Lungs deases → due to cotton

Black " " → cement / coal

Nasal Hair

Cilia → Air filter
(नाक के बाल)

mucus → Air temp.
(cough)